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Land revenues, schools and literacy: A historical examination of public and private funding of education

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Despite the centralised nature of the fiscal system in colonial India, public education expenditures varied dramatically across regions with the western and southern provinces spending three to four times as much as the eastern provinces. A significant portion of the inter-regional difference was due to historical differences in land taxes, an important source of provincial revenues in the nineteenth and early twentieth century. The large differences in public spending, however, did not produce comparable differences in enrolment rates or literacy in the colonial period. Nonetheless, public investments influenced the direction of school development and perhaps the long run trajectory of rural literacy.

Introduction

Although average levels of schooling and literacy in British India were low under the Raj, there were important regional differences in the development and progress of education.¹ In the Education Despatch of 1854 the East India Company

¹ British India refers to two-thirds of the Indian subcontinent that was under direct colonial control, whereas the remaining one-third of the territories known as 'Princely States' or 'Native States' were under the control of native kings that deferred to the British with regard to defence and foreign policy. In return, the native kings managed their local affairs such as education, infrastructure, etc., internally.

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established national guidelines to promote schooling, but the provinces responded to the suggestions in different ways.² Bombay ignored the former indigenous system of schools and constructed a large network of public schools, while Bengal relied on private funding to either build new private schools or incorporate indigenous schools into the new state system. Bombay went on to become a leader in public spending on education and Bengal, Bihar and Orissa fell further behind.

In this article, I first document important differences in enrolment and literacy rates between the Indian provinces. Then, I assess whether differences in public spending and school-systems influenced the observed enrolment and literacy patterns. Although this exercise answers some questions about educational outcomes and financing in British India, it also raises interesting questions for future work in the field. Despite the important relationship between education and economic development, the study of colonial education has received limited attention by economists and historians alike as compared to more traditional public investments such as railways. This study is a first step and I hope it will lead to more work on public finance and returns to education in the colonial period.

While the existing literature provides detailed commentaries and qualitative analysis of official policies, the interaction between education and nationalism and the influence of missionaries,³ regional differences in enrolments and public spending have largely escaped quantitative investigation although historians have commented on these patterns. Using a quantitative approach, I construct provincial series on enrolments, schools and spending from 1891 to 1931 and discuss how colonial land tax policies may have influenced provincial school systems.

Both enrolment and literacy were higher in the coastal provinces of Bengal, Bombay and Madras as compared to the interior provinces of Central Provinces and United Provinces. The coastal provinces had big urban cities with larger bureaucracies such as Bombay and Calcutta that perhaps offered more opportunities for educated workers positively influencing both the private demand for education and subsequent outcomes. Public investments, however, were an important part of the story although they did not produce comparable effects on outcomes.

Under the Raj, both public and private expenditures accounted for roughly 50 per cent of total education spending at the national level, although the composition of spending differed between provinces. From 1881 to 1931, Bombay led

² Given the high level of illiteracy in this period, the Despatch promoted the idea of allocating partial public subsidies to schools under private management as opposed to developing an extensive network of government funded and managed schools.

³ See Nurullah and Naik, *History of Education*; Basu, *Essays*; and Bellenoit, 'Missionary Education'. Researchers have also explored the influence of Western and English education, and inter-caste plus inter-religious differences in educational development. This is far from an overview of the topics covered, but is a sampling of the more popular areas of study in the literature. See Whitehead, 'Historiography of British Imperial Education Policy' for a systematic overview.

the way in public expenditures developing a large network of public schools. But public spending in Bengal, Bihar and Orissa lagged behind both Bombay and Madras.

A large share of the difference was due to land tax revenues. Bombay and Madras had higher land revenues on average and hence they had more public money available to spend on education (and other local services) as compared to Bengal where land revenues were lower on account of the Permanent Settlement. The Settlement was a contract between the English East India Company and the landlords of Bengal and Bihar whereby the revenue demand on land (land tax) was fixed in cash for perpetuity in 1793. In comparison, Temporary Settlement areas such as Bombay and Madras were assessed land taxes at higher rates that were periodically adjusted to account for changes in price levels and productivity. State capacity captured by public educational spending, thus, varied with the land tax regime and jointly influenced the development of schooling along with private spending on education.

Although public revenues were lower in Bengal, private revenues in the form of fees accounted for a larger share of total spending as compared to Bombay (40 per cent in Bengal versus 20 per cent in Bombay). This is also reflected in primary school fees that were higher in the public schools of Bengal as compared to Bombay (Rs. 2.4 per pupil and Rs. 0.7 respectively in 1901–02).⁴ Bombay used the higher public revenues to construct new public schools charging lower fees, while Bengal developed a system of private schools and high fees.

Although the composition of spending influenced the direction of school development (public or private), they had minor effects on aggregate literacy at least in the colonial period. For example, the big difference in literacy was not between Bengal and Bombay, rather the difference was notable between Bengal and the United Provinces or Bombay and the Central Provinces. However, a heavier reliance on private funding may have produced a more unequal school system with long run consequences for the development of literacy. Bengal had the largest proportion of English literates to total literates in the colonial period and large differences in rural literacy developed between Bengal and Bombay by 1961. The rest of the paper is organised as follows: the next section documents enrolment and literacy patterns; section 3 explores provincial differences in education spending; section 4 studies the implications of public spending on schools; section 5 discusses the relationship between investments and outcomes; and section 6 concludes.

⁴ Statistics from Progress of Education in India 1897–98 to 1901–02: Fourth Quinquennial Review.

Enrolments and Literacy

A systematic enumeration of literacy and enrolment rates in British India began in the late nineteenth century after the Hunter Education Commission Report published in 1883 noted the importance of collecting education data and statistics. In 1886–87, the first Quinquennial Report on the Progress of Education in India was released and subsequent Quinquennial Reports were published up to 1947 reviewing the development of education for each five-year period. Data in these reports refer to recognised institutions, while unrecognised indigenous institutions such as makhtabs and tols are often excluded because they were frequently opening and closing. By the early twentieth century, indigenous unrecognised education had all but disappeared in British India accounting for less than 10 per cent of schools and pupils. The new state system of schools introduced in 1854 dominated the education sector.

Using the Quinquennial Reports, Table 1 presents enrolment rates in recognised institutions from 1891 to 1931 at the primary, secondary and collegiate level for British India and the main Indian provinces. Total enrolment increased from less than 10 per cent of the school age population in 1891 to 30 per cent by 1931. The big increases in enrolment, however, occurred at the secondary and collegiate level as opposed to the primary level. College enrolment increased almost five-fold and accords with historical accounts of Indians flocking to post-secondary institutions in large numbers.⁵ In fact, India had a larger number of secondary and post-secondary students than several industrialised countries such as France even though primary school enrolment rates were among the lowest in the world.⁶

However, overall enrolment rates mask the substantial heterogeneity within British India. At every level, enrolment rates in the coastal provinces of Bengal, Bombay and Madras exceeded those in the interior provinces of Bihar and Orissa, Central Provinces and United Provinces. By 1931, almost 40 per cent of the children in the coastal provinces were enrolled in some school, compared to 20 per cent of the children in the interior provinces. Enrolment patterns also varied within the coastal provinces. For many decades, Bengal enjoyed the highest rates of secondary and collegiate enrolment although Bombay moved ahead by 1931. In comparison, primary school enrolment was always lower in Bengal than Bombay or Madras. Part of the difference is an artifact of the data, because middle–vernacular students are included with primary school students in Bombay and Madras, but with secondary students in Bengal.⁷ Adjusting for this difference, Bombay and Madras still enjoyed higher primary school enrolment. Since Bengal was the seat

⁵ For example, Basu, *Growth of Education*.

⁶ Chaudhary, 'Determinants of Primary Schooling'.

⁷ Secondary schools in this context often had attached primary classes and were in some sense an imperfect substitute to a vernacular primary school.

Table 1
Enrolment Rates (Pupils/School-Age Population Reported in Percentages)

	1891-92	1901-02	1911-12	1921-22	1931-32
All Recognised Institutions:					
<i>British India</i>	9.6	10.8	16.0	20.9	30.2
Bengal	12.7	13.5	24.4	26.2	36.2
Bihar and Orissa			15.5	15.0	18.4
Bombay (inc. Sind)	14.0	14.9	20.7	30.9	39.7
Central Provinces & Berar	7.0	8.9	13.4	15.8	19.4
Madras	11.8	12.9	18.6	27.5	41.0
Punjab	4.8	6.0	10.6	17.8	33.9
United Provinces	3.1	5.2	8.8	14.2	20.1
Primary Schools:					
<i>British India</i>	8.1	8.9	13.0	17.0	23.6
Bengal	10.7	11.0	17.7	20.5	28.2
Bihar and Orissa			13.0	13.5	15.6
Bombay (inc. Sind)	12.8	13.4	18.6	27.5	34.9
Central Provinces & Berar	5.7	7.4	10.9	12.5	14.3
Madras	10.3	10.9	16.6	24.4	37.4
Punjab	3.2	3.7	7.0	10.3	13.6
United Provinces	2.2	4.1	7.2	12.2	16.5
Secondary Schools:					
<i>British India</i>	1.4	1.7	2.4	3.3	5.7
Bengal	1.8	2.2	5.1	4.7	6.0
Bihar and Orissa			1.2	1.2	2.3
Bombay (inc. Sind)	1.1	1.3	1.8	2.7	3.8
Central Provinces & Berar	1.2	1.4	2.4	3.2	4.8
Madras	1.4	1.8	1.8	2.7	2.9
Punjab	1.6	2.1	3.3	7.2	19.2
United Provinces	0.8	1.0	1.4	1.6	3.0
Arts and Professional Colleges:					
<i>British India</i>	0.0	0.1	0.1	0.2	0.2
Bengal	0.1	0.1	0.2	0.3	0.3
Bihar and Orissa	0.0	0.0	0.0	0.0	0.1
Bombay (inc. Sind)	0.0	0.1	0.1	0.3	0.4
Central Provinces & Berar	0.0	0.0	0.0	0.0	0.1
Madras	0.1	0.1	0.1	0.2	0.2
Punjab	0.0	0.1	0.1	0.2	0.4
United Provinces	0.0	0.0	0.1	0.1	0.2

Source: Progress of Education in India, Quinquennial Reviews (1891-1947). Data are not shown individually for the smaller provinces and administrations. Up to 1911/12, the data include some native states of Bombay, Bengal, Central Provinces, & United Provinces but they represent 5% of the covered population. Secondary schools include high schools, middle English and middle vernacular schools. However, primary schools include middle vernacular schools for Bombay (all years) and for Madras from 1911-12. Bihar and Orissa are included in Bengal from 1891-1901. Enrolment and population data for Bengal, Bihar and Orissa are from the Statistical Abstracts for 1911/12. School-Age Population is defined as 15% of the total population.

of colonial administration until 1911, the patterns may partially reflect differential returns to secondary education among the coastal provinces.

Enrolment captures one dimension of educational development, which is perhaps more correlated with the availability of schools and spending. But it is far from a satisfactory measure because it is difficult to interpret enrolment as educational performance in the Indian context. Students may have enrolled in school, but if they did not learn anything, then enrolment rates are not a good statistic of achievement. The breakdown of schools into primary and secondary also varies over time and across province, making across-province comparisons difficult. Finally, recognised private schools were more likely to over report their enrolments because public subsidies were often tied to enrolment. Given these concerns, literacy is perhaps a more accurate measure of educational progress. Beginning in 1911, the Indian censuses began to enumerate literacy in a consistent and uniform manner across provinces. An individual with the ability to read and write a short letter to a friend was recorded as literate and official opinion indicates the definition was closely followed.

Table 2 reports total literacy, literacy in the population aged 10 to 20 and English literacy in the top panel and the three measures for males only in the bottom panel.⁸ Literacy was extremely low in early twentieth century India, averaging less than 10 per cent across the large Indian provinces. Surprisingly, literacy rates do not accord with the increase in enrolment rates observed in Table 1, especially in the 1920s. Total enrolment increased from 20 to 30 per cent between 1921 and 1931 without a parallel increase in literacy. If anything, literacy rates are stagnant with the number of literates in each decade barely keeping pace with the increase in population. Total literacy increased from 5.9 in 1921 to 6.6 per cent in 1931. Public officials were well aware of the discrepancy between enrolment and literacy, and various reports emphasise the inefficiency and wastage in the education system.⁹ As spending increased in the twentieth century, the total number of schools

⁸ There is a potential under enumeration of literates in Bengal in the 1931 census because of an absolute decline in 1931 male literacy rates relative to 1921. Official discussions attribute the decline to an under enumeration of literates in the 5 to 10 aged cohort, but this is an unsatisfactory explanation because there is an absolute decline even for the 20 and over age cohort in a few districts. Moreover, the official explanation for under enumeration should apply to females as well as males and there is no absolute decline in the number of female literates in the younger cohorts. Although official accounts are dismissive of an economic explanation, the general economic depression in the 1920s was particularly harsh in Bengal and may have contributed to relapses into illiteracy. In fact, the Quinquennial Reviews (Richey, J.A. 1924 and Littlehailes 1929) frequently discuss problems of wastage and frequent relapses into illiteracy.

⁹ The Tenth Quinquennial Review for 1927–32 has the following quote from the Hartog Committee on education: ‘Throughout the whole educational system there is waste and ineffectiveness. In the primary stage, which from our point of view should be designed to produce literacy and the capacity to exercise an intelligent vote, the waste is appalling.’

Table 2
Literacy Rates (in percentages)

	Overall			Ages 10 to 20			English			English literates as % of total literates		
	1911	1921	1931	1911	1921	1931	1911	1921	1931	1911	1921	1931
Bengal	7.7	8.5	8.6	9.2	9.9	9.6	1.1	1.3	1.7	14	15	20
Bihar & Orissa	4.1	4.7	4.5	4.8	5.2	5.1	0.2	0.4	0.5	6	8	10
Bombay	6.1	7.4	8.2	8.0	10.3	10.3	0.6	0.9	1.1	10	12	13
Central Provinces & Berar	3.6	4.6	5.6	5.3	6.6	7.0	0.3	0.5	0.6	9	10	11
Madras	7.5	8.6	9.5	8.7	10.0	10.9	0.7	0.9	1.3	9	11	14
Punjab	3.9	4.2	5.3	4.1	5.1	6.5	0.5	0.7	1.1	14	18	20
United Provinces	3.4	3.7	4.7	4.3	4.7	5.7	0.3	0.4	0.5	8	10	12
Average	5.2	5.9	6.6	6.3	7.4	7.9	0.5	0.7	1.0	10.1	12.0	14.1
Male												
Bengal	14.0	15.0	14.6	15.9	16.6	15.8	2.0	2.4	3.1	14	16	21
Bihar & Orissa	8.0	8.8	8.4	8.5	9.0	8.9	0.4	0.7	0.8	5	8	10
Bombay	10.9	12.7	14.0	13.0	15.8	16.5	1.1	1.5	1.8	10	12	13
Central Provinces & Berar	6.8	8.4	10.2	9.3	11.2	12.2	0.6	0.8	1.1	9	10	10
Madras	13.8	15.2	16.6	14.6	16.0	17.4	1.2	1.7	2.3	9	11	13
Punjab	6.5	6.9	8.5	6.2	7.8	10.0	0.9	1.2	1.8	14	18	21
United Provinces	6.1	6.5	8.0	7.0	7.5	9.3	0.5	0.4	0.9	8	10	12
Average	9.5	10.5	11.5	10.6	12.0	12.9	1.0	1.3	1.7	9.9	12.2	14.3

Source: Provincial Volumes of the Census of India for 1911, 1921 and 1931. Literacy is measured by crude literacy rates (literates/total population).

increased rapidly, but often these new village schools were inadequately staffed, with poor quality teachers.

Besides the low overall level, there were important regional differences. Madras enjoyed the highest literacy rate with Bengal and Bombay following close behind. Literacy mirrors the patterns on primary school enrolment in Table 1, where Madras also exceeds Bengal and Bombay. Similar to enrolment, literacy in the coastal provinces was almost twice as high as the interior provinces. Among the interior provinces, Punjab stands out as having a particularly inefficient education sector. In the 1920s, secondary and collegiate enrolment more than doubled while literacy in the population aged 10 to 20 only increased by 26 per cent.

Although literacy was low in India, the small numbers of literates were highly educated. More than 10 per cent of them could read and write in English, a second language, even though a vast majority of the people were unable to read and write in any language, starkly illustrating the unequal nature of the education system. Many of the English literates belonged to the upper castes and the new urban intelligentsia. In *Village, Caste, Gender and Method*, M.N. Srinivas argues that the new opportunities from western education 'had the twin effect of increasing the cultural and ideological divide between the high and the low castes, as well as making the new opportunities doubly desirable. In the first place, they were well paid and prestigious, and in the second, only the high castes had access to them (p. 78).' This inequality is especially evident in the case of Bengal. By 1931, male English literacy in Bengal averaged 3.1 per cent and 21 per cent of male literates were literate in English as compared to 13 per cent in Bombay and Madras.¹⁰

What can explain these varying patterns of educational development within British India? A simple economic model suggests individuals invest in education when the returns or benefits (i.e., higher wages) exceed the costs (i.e., school fees, opportunity cost in terms of lost wages). Market failures, however, can break down this simple model if people lack information on costs and benefits, if credit markets fail to provide loans to poor people to finance their education and if the social returns to education exceed the private returns. Such market failures often lead to less than optimal investments in education and states intervene in the education market by providing public schools or by allocating public money to private schools. Public spending can increase enrolments if public funds are used to provide schools and lower fees. Higher enrolments in turn can increase literacy if children learn to read and write in schools. One can think of public spending determining the public supply of schools, while wages and other factors related to returns influence the private demand for education. Although theoretically we can view supply or public spending as separate from demand, in practice it is more complicated because areas with high demand for education may also be in the position

¹⁰ We should interpret this with caution given the measurement error in the 1931 male literacy estimates for Bengal.

to raise more public funds and vice versa. This simple economic framework can partially explain the stark differences in enrolment and literacy between the coastal and interior provinces.

Economic opportunities for literate workers were perhaps more readily available in coastal provinces where the three major urban port cities of Bombay, Calcutta and Madras were located. Private demand for education is generally influenced by a region's level of development, crudely captured by income and urbanisation rates. In the Indian context, Bombay and Madras had above average urbanisation rates (19 and 12 per cent respectively in 1911) but so did United Provinces and Punjab (10 and 11 per cent respectively). So, private demand for education was perhaps more related to the size of the bureaucracy in the bigger urban cities and the opportunity for non-traditional employment in law, state offices, etc. These alternate measures represent another facet of development, separate from income and urbanisation.

But how does public spending fit into the picture? What explains the educational differences between Bengal, Bombay and Madras? Did private demand determine public financing of education across provinces? Or did the colonial state provide more funds to less developed provinces? To answer such questions, the next two sections study educational expenditures and school systems under the Raj in more detail.

Educational Expenditures

The East India Company largely ignored the subject of education for many decades, but beginning in the early nineteenth century it began to take a more active interest culminating in Wood's Despatch of 1854, which outlined a national education system for British India. Following the Revolt, the British Crown adopted the general guidelines of the Despatch and controlled education policy until 1919 relying on advice from both the Governor-General of India and the various provincial governors. Over this period, the former indigenous schools of India were either incorporated into the new system or disappeared due to lack of patronage.¹¹

Under the new system, education was funded from provincial government revenues, local government funds (rural district and urban municipal councils), fees and other private contributions such as endowments. In the mid-nineteenth century, the central government had absolute fiscal authority, but starting in the 1870s the provision of education was first decentralised to provincial governments and then

¹¹ Adam, *Third Report on the State* (1838) conducted a detailed enquiry on the state of indigenous education in Bengal in the early nineteenth century. His reports suggest indigenous schools were remarkably prevalent. Families would pool their resources to provide money and school accommodations for a teacher to educate their boys. His rough calculations suggest 5 to 6 per cent literacy for parts of Bengal, which seem a bit high but the literacy standard (ability to decipher or sign name) was lower than used for the colonial censuses (ability to read and write a short letter or paragraph).

further decentralised to lower levels of government such as rural district and urban municipal boards in the 1880s. In general, the local boards managed primary education, while secondary and collegiate education came under the control and supervision of provincial governments.

Over the late nineteenth and twentieth century, provincial governments contributed approximately 30 per cent to total revenues, district and municipal governments accounted for 16 per cent and 4 per cent respectively, fees were another 33 per cent, and private contributions made up the rest. Although expenditures increased over this period, the big increases came after 1919 when the Montague–Chelmsford reforms ushered in the period of dyarchy. Certain categories of revenue such as land revenues were completely transferred to provincial governments that now included elected Indian ministers and education became a transferred subject under the control of provincial legislatures. With the introduction of Provincial Autonomy in 1935, provincial legislatures were given more fiscal independence and educational investments increased again.¹² Given the high costs of building a mass education system, the colonial government initially relied on public and private revenues.¹³ Beginning in the 1850s, the government actively encouraged private schools and also began building a network of public schools, funded and managed either by provincial governments or local boards (district and municipal). To promote private provision, public subsidies in the form of ‘grant-in-aids’ were offered to privately managed schools to defray part of their operating expenditures.¹⁴ These schools came to be known as aided schools and functioned alongside unsupported private unaided schools as well as pure public schools (i.e., publicly funded and managed). Although public spending on average accounted for half of total spending on education, it grew to almost 60 per cent of total spending by Indian independence in 1947.¹⁵

Table 3 on nominal educational expenditures per-capita reflects the division between public and private spending from 1881 to 1931. Before we interpret the patterns, two peculiarities of the data should be noted. First, the official data include information on a small number of Princely States for 1901 and 1911. The data,

¹² See Chand, *The Essentials*, Misra, *Indian Federal Finance*, and Kumar, *Cambridge Economic History* for details on the colonial fiscal system.

¹³ See Nurullah and Naik, *History of Education*, Basu, *Essays* and Ghosh, *History* for details on colonial education.

¹⁴ Under this system, public grants were made available to private schools known as aided schools that followed a secular curriculum, were under local management and open to official inspection. Grants were often allocated to specific expenditures such as teacher salaries and covered about a third to half of the total operating expenditures of the school. Provincial governments were given substantial leeway in developing grant eligibility rules and there was substantial heterogeneity across provinces in the development of aided schools.

¹⁵ Progress of Education in India, Twelfth Quinquennial Review (1937–47).

nonetheless, are fairly representative of spending in British India because the Princely States in the series account for less than 5 per cent of the population.¹⁶ Second, total expenditures are disaggregated into spending from public revenues and private revenues such as fees and endowments. But expenditures from public revenues in Table 3 may understate public expenditures because public schools often charged fees that are included under expenditures from private revenues.

Despite the caveats, Table 3 shows total spending increased dramatically from 1881 to 1931—expenditures in 1931 were almost ten times higher than in 1881. Over these decades, prices increased anywhere from a low of 22 per cent to a high of 59 per cent depending on the price index employed.¹⁷ Spending thus increased in real terms though British India still had lower expenditures on human capital as compared to both Indian Princely States and other underdeveloped countries.¹⁸ Moreover, total spending on education in 1931 accounted for less than 0.5 per cent of income per-capita.

Table 3 also highlights the across province variation in total expenditures per-capita, which mirror the patterns on expenditures from public revenues. Bombay represents one extreme with the highest provincial spending per-capita in each decade, while Bengal and Bihar and Orissa are at the other lower extreme of spending. The differences are nontrivial with Bombay out spending the eastern provinces by a factor of 4. Bengal with its large state bureaucracy in Calcutta perhaps generated a higher private demand for schooling and so private revenues compensated for the low levels of public spending, but Bihar and Orissa lacked comparable opportunities for educated workers, so low public spending went hand in hand with low private spending.

Most of the regional variation in public spending was due to historical circumstances. Differences in provincial revenues date back to the mid-nineteenth century when the fiscal system was extremely centralised (1833–70) and the central government allocated funds to provincial governments. Till 1871, provincial governments of British India had no financial independence—all revenues accrued to the central government, which then disbursed funds to provinces based on their estimates of need.

Historians and contemporaries of the period argue that the allocation of funds was extremely random without any regard to the underlying economic situation

¹⁶ The 1901 and 1911 series from the Quinquennial Reviews include data on the princely states of Bombay (almost all of them), Madras (1), Bengal (1), and Central Provinces (several). Overall, data on princely states represent 5 per cent of the population covered. Bengal includes data for Bihar and Orissa pre-1911.

¹⁷ Calculations based on index of Indian prices reported in Appendix Table 11 A.1 in Kumar, *Cambridge Economic History*.

¹⁸ See Davis and Huttenback, *Mammon and the Pursuit of Empire*.

Table 3
Educational Expenditures per 1,000 of the Population

	1881–82	1891–92	1901–02	1911–12	1921–22	1931–32
Total Expenditures						
<i>British India</i>	95	139	167	211	744	1,000
Bengal	96	131	160	384	715	844
Bihar & Orissa				170	339	462
Bombay	207	332	306	397	1,530	1,832
Central Provinces & Berar	79	97	108	147	584	669
Madras	112	165	208	228	803	1,214
Punjab	84	128	160	214	917	1,307
United Provinces	46	71	96	141	657	804
Expenditures from Public Revenues						
<i>British India</i>	53	71	79	107	466	620
Bengal	40	51	51	130	325	351
Bihar & Orissa				82	217	283
Bombay	122	169	186	255	1,115	1,226
Central Provinces & Berar	65	74	77	117	483	494
Madras	53	78	82	97	458	742
Punjab	65	87	88	118	590	870
United Provinces	36	45	55	89	440	558
Expenditures from Private Revenues (Fees, Endowments, etc.)						
<i>British India</i>	42	68	87	104	279	381
Bengal	56	80	109	254	390	493
Bihar & Orissa				88	122	179
Bombay	85	163	120	143	415	606
Central Provinces & Berar	14	24	30	30	101	176
Madras	59	87	125	131	345	472
Punjab	19	41	72	96	327	438
United Provinces	11	27	40	52	217	246

Source: Statistical Abstracts for 1881 and 1891 series. Progress of Education in India, Quinquennial Reviews (from 1897–1902 to 1927–1932) for 1901 to 1931 series. Bihar and Orissa are included under Bengal from 1881–1901. 1911 data for Bengal, Bihar and Orissa are for 1916–17 due to reporting issues with the 1911 province boundaries. The 1901 and 1911 series also include data on some princely states but they only represent 5% of the population covered in the reviews. All figures are in nominal rupees.

of the province.¹⁹ According to Chand in *Essentials of Federal Finance*, the provincial ‘distribution of funds was not and could not be effected according to any fixed or intelligible principle, and the expenditure of the Provinces was determined, not by their resources or requirements, but by the degree to which they

¹⁹ See Chand, *The Essentials* and *Some Aspects* for details. Misra, *Indian Federal Finance* and Kumar, *Cambridge Economic History* have made similar arguments based on statistics reported in Chand, *The Essentials*. Sir Richard Strachey, a British administrator, and Mr. Gokhale are two contemporaries of the period that are often quoted with regard to this issue. According to Sir Richard Strachey,

were claimant on pressing their demands on the Government of India' (p. 34). This period of extreme centralisation led to disparities in inter-provincial public spending with Bombay emerging as one of the highest spending provinces and Bengal among the lowest spending provinces by 1870.

The inequities in public expenditures strengthened over the late nineteenth century as financial decentralisation was gradually introduced.²⁰ In the first stage of decentralisation (1871–77), the central government used extremely unequal 1870 expenditures to set the amount of fixed grants allocated to provincial governments in 1871. Subsequent resettlements often used past expenditures as a guide and no attempts were made to balance spending differentials across provinces. In fact, the distribution of revenues often exacerbated existing differentials with Bombay receiving larger per-capita grants as compared to Bengal.²¹ Under dyarchy (1921–35), Indian ministers increased public investments on education but the new system did not alter the regional patterns even though Bombay lost an important source of revenue namely income taxes that were now credited to the central government. Since Bengal suffered the same fate, the regional inequities remained unchanged. Colonial policies tried to reduce the differences, but were largely ineffectual and as late as 1931, Bihar and Orissa was the lowest spending province.

Bombay and Bengal thus inherited varying public revenues in 1871 on account of historical anomalies linked to the centralisation period. Beginning in 1882, however, differences in land revenues accounted for bulk of the variation in spending because provinces began to receive a share of their land revenues that were earlier credited to the central government. Regional differences in land tax revenues

'the distribution of public income degenerated into something like a scramble, in which the most violent had the advantage, with very little attention to reason' (Chand, *The Essentials*, p. 34). In testimony before the Welby Commission in 1896, Mr. Gokhale indicated that 'The fact is that these inequalities are a legacy from the pre-decentralization period, when the expenditures of different Provinces was determined ... not by the resources or requirements of these Provinces, but by the attention that their Governments succeeded in securing from the Central Government i.e. by the clamour that they made' (Misra, *Indian Federal Finance*, p. 37).

²⁰ Pre-1870, the central government made annual assignments or grants to the provincial governments for each category of expenditure. After 1870 provincial governments received fixed assignments (as total revenues from certain categories such as jails, police, education and medical services and either a fixed grant till 1881 or a share of the land revenue after 1882) and they could allocate expenditures as they chose. In practice high spending provinces spent above average on all categories of provincial expenditures. See Chand, *The Essentials* for details.

²¹ As part of the 1887 Quinquennial Settlement, provinces were given a share of the land revenue raised in their respective provinces. Initially Bombay was granted 60 per cent of the land revenue, Bengal received 32 per cent. Moreover, the proportions underestimate the extent of the differential because on account of the Permanent Settlement, land revenues in Bengal were significantly lower than in Bombay. In subsequent settlements, the proceeds from land revenue were equally shared between the provincial and central government. See discussion in Kumar, *Cambridge Economic History*.

(or land revenues) were directly related to the type of Land Settlement negotiated by the British in different parts of the subcontinent.

In Bengal and Bihar, the East India Company came to an agreement with the landlords in 1793 whereby they were made responsible for tax payments and land taxes were fixed in cash for perpetuity.²² No surveys of agricultural or economic conditions were conducted and revenue amounts were largely a function of bargaining between the two parties and of historical taxes under the Mughals.²³ This arrangement came to be known as the Permanent Settlement of 1793. On account of this Land Settlement, Bengal and Bihar were unable to recover higher revenues as agricultural productivity and prices increased in the nineteenth and twentieth century. The colonial government also had fewer incentives to increase public investments in these areas because they were unlikely to recover any of the gains due to the fixed nature of the revenues. In comparison, land revenues were negotiated with individual cultivators or village bodies, and revised every thirty years in non-Permanent or Temporary Settlement areas. Detailed land surveys were employed at each reassessment to establish the rightful 'land owners' or parties responsible for the tax payment and to determine the assessments that were then taxed at variable rates depending on the province. Tax rates were higher in the southern and western provinces but declined uniformly over the late nineteenth and twentieth century. By 1930 land revenues captured less than 5 per cent of gross agricultural produce.²⁴ Assessments were a function of soil type, rainfall and certain socio-economic conditions. The system, however, was far from perfect and subjective forces often came into play.²⁵

The top panel of Table 4 reports per-capita land revenues collected in British India and the larger provinces between 1881 and 1931. Differences between the Permanent and Temporary Settlement provinces are very stark. Bombay contributed over 2 rupees per person on average, while Bengal contributed 0.5 rupees. Bengal had more fertile alluvial soil than Bombay and so differences in land revenues between the two are not proxies for agricultural productivity. Since revenues were static in Permanent Settlement Bengal and Bihar, they remained at their 1793 level whereas they were adjusted every thirty years in the Temporary Settlement provinces. Provinces with above average land revenues also had higher public spending, as seen in Table 3.

²² See Baden-Powell, *A Short Account of the Land Revenue*, Kumar, *Cambridge Economic History* and Banerjee and Iyer, 'History, Institutions and Economic Performance' for details on land tenure.

²³ See Baden-Powell, *The Land Systems of British India* (1972) for details on the Permanent Settlement.

²⁴ See Kumar, *Cambridge Economic History*, p. 918.

²⁵ Chand, *The Essentials*, Chand, *Some Aspects of Fiscal Reconstruction and Report of the Indian Taxation Enquiry Committee*.

Table 4
Land Revenues and District Board Revenues per Capita

	1881–82	1891–92	1901–02	1911–12	1921–22	1931–32
Land Revenues						
<i>British India</i>	1.10	1.08	1.18	1.28	1.41	1.20
Bengal	0.57	0.54	0.55	0.37	0.65	0.61
Bihar & Orissa					0.48	0.47
Bombay	2.33	2.35	2.38	2.41	2.83	2.29
Central Provinces & Berar	0.62	0.65	0.88	1.34	1.90	1.36
Madras	1.41	1.11	1.33	1.37	1.44	1.14
Punjab	1.09	1.11	1.23	1.45	1.19	0.94
United Provinces	1.30	1.26	1.33	1.36	1.50	1.27

Source: Statistical Abstracts relating to British India. Bihar and Orissa are included under Bengal up to 1911–12. Bengal in 1911–12 excludes Eastern Bengal due to the partition of Bengal.

District Board Revenues						
	<i>Cess—additional surcharge on land revenues</i>			<i>Provincial grants</i>		
	1895–96	1909–10	1920–21	1895–96	1909–10	1920–21
Bengal	0.05	0.05	0.14	0.01	0.02	0.05
Bihar & Orissa			0.19			0.02
Bombay	0.15	0.17	0.17	0.06	0.13	0.31
Central Provinces & Berar	0.04	0.08	0.06	0.01	0.05	0.19
Madras	0.15	0.16	0.18	0.01	0.04	0.16
Punjab	0.11	0.14	0.21	0.00	0.07	0.21
United Provinces	0.04	0.08	0.16	0.02	0.05	0.12

Source: Gyan Chand, *Local Finance in India* (1947), tables Appendix VI, Appendix X and Appendix XII. Population data was obtained from the Statistical Abstracts. Chand (1947) is unclear whether Bengal in 1909–10 excludes Eastern Bengal, which was a separate province merged with Assam over these years on account of the partition of Bengal in 1905. I have used the population figures for Bengal Proper, Bihar and Orissa but this could generate some mis-measurement in the Bengal per-capita numbers for 1909–10 if Chand (1947) is using figures for Bengal excluding Eastern Bengal.

Demographic differences in population growth further reinforced the differences in the per-capita patterns of land revenues. Between 1780 and 1940, Bombay suffered multiple famines (1876–78, 1896–97 and 1899–1900) with high mortality, while Bengal suffered no serious famines.²⁶ Since land taxes were fixed by land area and not by population, the droughts and famines of Bombay ensured the larger absolute level of revenues were distributed over a stable population, keeping per-capita revenues high for most of the period. In comparison, the lower land

²⁶ Bihar did suffer from a famine in 1873–74 but it did not affect mortality because of generous government support.

revenues in Bengal were distributed over a growing population base leading to low and possibly declining per-capita revenues.²⁷

Land revenues also determined spending by lower levels of government because rural district boards financed the provision of local services such as primary education via additional surcharges (of around 6.25 per cent) on land revenues.²⁸ As seen in the bottom panel of Table 4, these surcharges known as cesses are strongly related to land revenues. Bombay with high land revenues also collected high cesses. In addition to cesses, provincial governments contributed money to local boards in the form of education grants. Again, provinces with access to more land revenues could allocate bigger grants to their districts. The bottom panel of Table 4 reports the amount of grants in 1895, 1909 and 1920. By 1920, Bombay district boards enjoyed the largest disbursements from their provincial government as compared to boards in Bihar and Orissa.

Thus, land revenues influenced public educational spending and in turn total educational spending. But, an important question is one of magnitude—how far can land revenues explain the inter-provincial differences in total investments on education? On average, private revenues (fees, endowments, etc.) accounted for almost 50 per cent of total spending on education under the Raj according to the Progress of Education in India, Quinquennial Review (1927–32). Among public revenues, land revenues determined at most 70 per cent of district board spending, assuming all the variation in cesses and grants was due to land revenues. But we still need to incorporate the effect of land revenues on total provincial revenues. A crude calculation from the Statistical Abstracts suggests land revenues contributed probably 60 per cent on average to total provincial revenues. Putting the two together, land revenues can approximately explain 32 per cent of total spending on education.²⁹

This very rough calculation assumes land revenues did not influence the private demand for education and hence private spending on education. If higher land revenues increased private spending on education, then land revenues would have a greater effect on total spending and vice versa, if higher land revenues reduced private spending on education. Given that private educational expenditures in

²⁷ I thank Tirthankar Roy for highlighting this point.

²⁸ Rural district boards were constituted by Lord Ripon in 1882 and were primarily responsible for the management of primary education, local infrastructure and medical services in rural areas. See Chand, *Local Finance in India* and Tinker, *Foundations of Local Self-Government* for details on local boards. In the Permanent Settlement Provinces, the cess was assessed on annual surveys that were supposed to capture the rent paid by a tenant to a landlord. It is unclear if these surveys were even conducted and in practice the correlation between cesses and land revenues in Permanent Settlement districts is very large.

²⁹ According to the Progress of Education, Tenth Quinquennial Review (1927–32), district board revenues accounted for 20 per cent of total educational expenditures and provincial revenues accounted for 30 per cent in the early twentieth century.

Bengal was very high, it is unlikely the Permanent Settlement reduced private spending on education although it may have skewed it away from primary education. Hence, 32 per cent is probably an upper bound on the effect of land revenues on total educational investments with private spending probably driving the remaining 68 per cent.

Hence, almost one-third of total spending was due to historical circumstances. The Permanent Settlement coupled with a lack of famines ensured low levels of per-capita revenues for Bengal and private revenues emerged to fill the gap. In comparison, public spending in Bombay was higher than other provinces allowing it to successfully develop a large network of public schools because of the easy availability of public funds and a stable population.

Land Revenues and Schools

Differences in public spending also had important consequences for the development of school systems and public subsidies offered in primary schools. Table 5 reports the number of schools per 100,000 people by province and management type.³⁰ Similar to expenditures, the number of schools increased, doubling from 44 schools per 100,000 people in 1891 to 82 in 1931. Most of the increase was in public schools that tripled over this period although private aided schools also increased by 75 per cent. In comparison, the number of private unaided schools declined in the early twentieth century because several poor quality one-teacher primary schools were closed in Bihar and Orissa, and Madras in the 1920s.

Bengal and Bombay were again at the two extremes with other provinces falling somewhere in between. Bengal had the largest number of private aided and unaided schools, while Central Provinces at the other end had fewer than 3 private schools per 100,000 people in 1931 as compared to 31 public schools. The Central Provinces, Punjab and United Provinces generally followed the Bombay model and focused on developing public schools while Bengal and Bihar and Orissa followed the path of private school development. Madras fell somewhere in between the two extremes because public schools accounted for less than 20 per cent of total schools in the 1890s, but increased to over 40 per cent by the 1930s. On average, provinces with access to larger public revenues developed more public schools.

Either provincial governments or local councils managed public schools, but many different groups operated in the private sector. Christian missionaries were responsible for the early wave of private school development, but private schools managed by Indians far out numbered mission schools by 1882. For example, Indians managed 97 per cent of primary schools and 63 per cent of secondary

³⁰ Schools in this table include all recognised educational institutions namely primary schools, secondary schools and colleges that were recognised under the new state system of education introduced by the colonial government in the 1850s. Data on the small number of unrecognised schools, often religious in nature, are excluded.

Table 5
Schools by Management Type per 100,000 of Pop

	1891-92	1901-02	1911-12	1921-22	1931-32	1931-32 school size
Total Recognised Schools						
<i>British India</i>	44	44	53	70	82	54
Bengal	74	68	84	111	135	40
Bihar & Orissa				76	77	36
Bombay	35	37	50	69	73	81
Central Provinces & Berar	20	24	25	35	34	84
Madras	56	54	62	88	115	54
Punjab	11	15	22	38	51	100
United Provinces	11	16	24	41	49	62
Publicly Funded and Managed Schools						
<i>British India</i>	9	8	12	23	30	71
Bengal	1	0.5	2	8	10	57
Bihar & Orissa				4	11	59
Bombay	27	19	29	52	57	82
Central Provinces & Berar	12	12	19	31	31	86
Madras	10	8	12	26	48	61
Punjab	10	10	16	30	41	97
United Provinces	10	10	14	33	36	66
Privately Managed Aided Schools						
<i>British India</i>	26	26	32	41	46	45
Bengal	59	50	63	89	105	38
Bihar & Orissa				57	58	10
Bombay	8	8	10	16	15	79
Central Provinces & Berar	7	9	3	3	2	89
Madras	29	30	39	55	64	49
Punjab	1	4	5	7	8	116
United Provinces	1	6	9	7	12	51
Privately Managed Unaided Schools						
<i>British India</i>	9	8	8	7	6	42
Bengal	14	16	19	14	20	43
Bihar & Orissa				16	8	28
Bombay	1	1	1	1	1	79
Central Provinces & Berar	1	1	2	2	1	45
Madras	17	16	11	7	2	36
Punjab	0	1	1	1	1	92
United Provinces	0	0	1	0	0	63

Source: Statistical Abstracts for 1881 and 1891 series. Progress of Education in India, Quinquennial Reviews (from 1897-1902 to 1927-1932) for 1901 to 1931 series. Bihar and Orissa are included under Bengal from 1881-1911. The 1901 and 1911 series also include data on some princely states but they only represent 5% of the population covered in the reviews.

schools in 1881–82.³¹ On average, secondary schools managed by Indians were more common in Bengal, Bombay and Madras as compared to Punjab where majority of the private English schools in the 1880s were mission schools.

Missionaries also dominated among arts and professional colleges in the nineteenth century, but by 1901–02 Indians managed 42 colleges as compared to 37 by missionaries. Indians also controlled most of the secondary schools by this time.³² Official policies, for example the recommendations of the Hunter Education Commission Report, championed private Indian efforts over missionary activities and contributed to these changes. Among Indians, wealthy landlords and local kings (former rajas) endowed several schools along with reform movements such as the Arya Samaj that established many private schools (Dayanand Anglo-Vedic or DAV schools) especially in Punjab.

Student enrolment rates also mirror the distribution of public and private schools observed in Table 5. In 1891–92, almost 80 per cent of the students in Bombay were enrolled in public schools as compared to less than 10 per cent in Bengal.³³ While the share of public school enrolment increased over the twentieth century in keeping with the larger public resources devoted to education, the marked differences between Bombay versus Bengal, Bihar and Orissa are visible as late as 1931. Just over 15 per cent of pupils in the eastern provinces were enrolled in public schools in 1931.³⁴ Provinces with high land revenues and above average public investments on education, namely Bombay, Central Provinces and Madras, developed a larger number of public schools as compared to provinces with below average public spending.

While Table 5 indicates that Bengal, Bihar and Orissa, and Madras had the largest number of schools per-capita, they also had smaller schools with an average school size of 43 students as compared to 82 students in other provinces. The last column of Table 5 reports average school size in 1931 and the numbers are representative of school size in earlier decades. Smaller schools in this context should not be confused with better quality schools because the former were often one-teacher, multi-grade primary schools located in rural villages and lacking basic infrastructure. Bigger schools, in comparison, were housed in proper buildings and had multiple teachers with some sort of training.

³¹ Nurullah and Naik, *History of Education*, p. 260.

³² *Ibid.*

³³ Calculations based on information in the Progress of Education in India, Quinquennial Reviews.

³⁴ This discussion refers to total enrolment across all types of schools (primary, secondary and colleges). In general, enrolment in public schools was higher at the primary level as compared to the secondary level where unaided secondary schools were more prevalent. Private funds played a bigger role at the secondary and collegiate level in British India although the colonial government also set aside vast sums of money to develop universities and several high quality secondary schools.

By relying on private schools, the eastern provinces may have also excluded some poor children from primary schools because private aided schools charged higher fees on average as compared to public schools. Table 6 lists average annual fees and expenditures per-pupil in 1901 and 1931 for public, private aided and all primary schools. Bengal had the highest primary school fees for both public and aided schools even though expenditures on primary schools were about average. More than 50 per cent of 1901 per-pupil expenditures were recuperated from fees in this province. Bombay schools, in contrast, offered a large public subsidy because public school fees were low relative to expenditures and public funds covered the difference.

The increase in public spending from 1901 to 1931 was used to lower primary school fees in both public and aided schools. By 1931, fees covered only 2 per cent of annual expenditures. Bengal, nonetheless, continued a heavier reliance on fees as compared to the other provinces. The ratio of fees to expenditures averaged around 2 per cent for Bombay schools and 14 per cent for schools in Bengal, Bihar and Orissa. Due to lower public subsidies, that is, higher primary school fees, these schools were perhaps less accessible to poor classes of the rural population as compared to schools in other parts of the country offering higher public subsidies.

Public education investments strongly influenced the composition of public and private schools within British India. Bombay devoted its higher public revenues to building more public schools with lower fees and a larger proportion of trained teachers. Due to fiscal constraints, Bengal, Bihar and Orissa used fees to extend education and frequently operated lower quality schools staffed by a large number of untrained teachers. Even as public investments increased in the 1920s and 1930s across India, these areas were unable to completely give up their reliance on fees that covered a significant portion of their expenditures relative to any other province.

Investments and Educational Outcomes

The two preceding sections noted the influence of inter-province differences in public investments on school systems. But, did these differences in public spending translate into differences in enrolments and literacy within British India? As discussed in the third section, land revenues roughly contributed to one-third of the variation in total educational investments, while two-third was determined by private spending. Hence, even if we assume one rupee of public spending had the same effect on outcomes as one rupee of private spending, private spending on education would probably account for at least two-third of the variation in enrolments across provinces with land revenues making up the rest.

But private spending appears to play a more central role in accounting for differences in enrolment and literacy. Neither enrolment nor literacy in Bombay is

Table 6
Primary School Fees, Expenditures and Teachers

	Per-Pupil						Fee/Expenditures		Trained teachers	
	Average annual school fee			Annual expenditures						
	Public	Aided	All schools	Public	Aided	All schools	Public	Aided	All schools	% all schools
1901-02										
British India	0.61	1.20	1.06	5.15	3.17	3.77	12%	38%	28%	.
Bengal	2.39	1.56	1.58	4.01	2.84	2.72	60%	55%	58%	.
Bombay	0.69	1.44	0.81	7.12	7.11	7.13	10%	20%	11%	.
Central Provinces & Berar	0.41	0.88	0.53	4.00	6.93	4.21	10%	13%	13%	.
Madras	0.98	1.03	1.08	4.90	3.76	3.68	20%	27%	29%	.
Punjab	0.70	0.58	0.77	4.97	3.18	4.53	14%	18%	17%	.
United Provinces	0.34	0.27	0.28	3.65	2.12	3.18	9%	13%	9%	.
1931-32										
British India	0.27	0.97	0.67	10.96	5.89	8.35	2%	16%	8%	50%
Bengal	1.03	1.67	1.45	7.25	3.72	4.00	14%	45%	36%	27%
Bihar & Orissa	1.16	1.14	1.22	8.22	6.15	6.33	14%	19%	19%	48%
Bombay	0.27	1.95	0.50	17.46	15.46	17.18	2%	13%	3%	48%
Central Provinces & Berar	0.19	2.81	0.34	9.95	18.54	10.28	2%	15%	3%	58%
Madras	0.13	0.55	0.33	10.88	6.43	8.44	1%	9%	4%	61%
Punjab	0.23	0.59	0.30	10.38	7.58	9.82	2%	8%	3%	67%
United Provinces	0.30	0.22	0.36	8.31	4.88	7.83	4%	4%	5%	63%

Source: Calculated from information in Progress of Education in India 1902-07 and Progress of Education in India 1927-32. Bihar and Orissa are included under Bengal for 1901-02.

30 per cent higher than Bengal. In fact, the big difference in both enrolment and literacy is not between Bengal and Bombay, but rather between the coastal provinces and the interior provinces. To understand the relationship between land revenues, total educational investments and outcomes, this section examines how private spending may have interacted with public constraints. The exercise answers some questions, while posing other questions to motivate future work in the field. The case of Bengal is particularly interesting. With a large state bureaucracy in need of educated workers, Bengal probably generated a high private demand for literate workers. Even though public spending was unable to meet this demand, private funding partially offset low public spending. Consequently, Bengal had a similar number of literates as the other coastal provinces of Bombay and Madras with perhaps comparable bureaucracies and opportunities for educated jobs in law and commerce. This suggests that returns in Bengal may have been higher than other provinces to account for the high levels of educational achievement. However, it is unclear if high returns were related to land revenues or land tenure, and if yes through what channel?

In Bengal, private spending was also more skewed to secondary schools and colleges heavily favoured by Indian elites. Upper castes and landed elites often established private schools and had strong preferences for English medium secondary schools where they could send their children. In fact, the landed elites of Bengal and the Bengali intelligentsia opposed any official attempts to extend mass primary education over the late nineteenth and early twentieth century, often pushing the state to increase spending for secondary schools and colleges.³⁵ The zamindari system of land tenure may have further exacerbated the fiscal constraints created by the Permanent Settlement. By making landlords responsible for revenue payments and thereby granting them 'property rights', the zamindari system may have also consolidated elite power in local villages, allowing landed elites to constrain any meaningful transfers to the larger population.³⁶

Whether the zamindari system alone, without the Permanent Settlement, would have produced such a skewed system is an open question. In the absence of fiscal constraints, Bengali elites would have needed state support to use public funds to put in place an educational system better suited to their needs. Would the British

³⁵ According to Mandal, 'The Ideology and the Interests of the Bengali Intelligentsia', Bengali elites actively opposed Campbell's (Bengal Lieutenant Governor) policy of providing more public support for mass education: 'The Bengali Hindu bhadraloks, a newly emerging social group mostly consisting of persons belonging to higher Hindu castes, after imbibing the best fruits of western education were trying desperately in the second half of the nineteenth century to assume the social leadership of the Bengalis. This social group, ambitious and ruthless, wanted to keep the masses ignorant in order to maintain their leadership over them' (p. 84). Mukhopadhyay, *Mass Education in Bengal*, offers numerous examples of Bengali elites resisting attempts to expand mass education.

³⁶ See Banerjee and Iyer, 'History, Institutions and Economic Performance'.

Government have complacently gone along with their wishes? Would the colonial government of Lord Curzon, for example, have bowed down to Indian elites especially in the early 1900s when British officials were more suspicious of Indian secondary education in general? Such questions all point to the complex interactions between the zamindari system and private spending on education.

The high levels of English literacy in Bengal provide further evidence on the unequal nature of the education system. A disproportionately larger share of total literates in Bengal could read and write in English relative to either Bombay or Madras (see Table 2). English literacy was an urban phenomenon because students in rural primary schools were instructed in the vernacular. Thus, a heavy emphasis on urban English literacy may have adversely influenced the subsequent progress of rural literacy. Although Bengal and Bombay had comparable literacy in the colonial period, rural literacy diverged sharply between the two provinces following Indian Independence. In 1961, rural male literacy was 20 per cent in West Bengal (part of the former province of Bengal) and 38 per cent in the former districts of Bombay province. Did historical public investments contribute to these long run patterns on rural literacy? Greater public revenues, a network of rural public schools and low fees probably gave Bombay a unique historical advantage, while low public funds probably hurt Bengal.

Colonial public spending may have had long run consequences on literacy, but the effects on colonial enrolment and literacy were not very significant. In addition to state employment opportunities, the Bengal and Bombay presidencies had big cities that were commercial hubs for different industries (jute in Bengal and cotton in Bombay). This would lead to higher private spending on education and could account for the comparable enrolment and literacy rates across the two provinces despite the sharp differences in the composition of total spending. The case of Central Provinces and Bihar and Orissa also suggests that public spending alone cannot explain differences in outcomes. Both these provinces were commercially less developed with a smaller provincial bureaucracy, and therefore returns to education were probably lower too. But the Central Provinces had a larger pot of public money because land revenues were flexible as compared to Bihar and Orissa. Per-capita expenditures from public revenues in 1931 were Rs. 494 in Central Provinces compared to Rs. 283 in Bihar and Orissa (Table 1). Yet, these differences did not produce significant differences in either enrolments or literacy. The total enrolment rate in 1931 was 18.4 per cent in Bihar and Orissa, and 19.4 per cent in the Central Provinces.

Why did such large differences in public spending fail to produce comparable differences in educational outcomes? To many economists, this finding is perhaps no surprise because if private returns to education (i.e., wages) are low, then individuals are unlikely to enrol in schools or become literate. Should we conclude then

that the public supply of schooling alone could not improve educational outcomes under the Raj without a parallel increase in private spending on education? This is a very pessimistic reading of the data and ignores the potential long run consequences of public investments on outcomes.

A more nuanced interpretation suggests that higher wages to educated workers were perhaps more effective than public spending at improving educational outcomes under the Raj. For example, by 1931 enrolment and literacy in Bombay were 7 per cent and 10 per cent higher than in Bengal. If we believe the crude estimate of land revenues driving up to one-third of the differences in total educational spending, then outcomes in Bombay should have been at least 30 per cent higher than Bengal assuming the two provinces had comparable private returns to education. But if one rupee of private spending increased enrolment or literacy by more than one rupee of public spending (i.e., pure public investments), a discrepancy between the two elasticities could explain why differences in public spending did not create large differences in outcomes. This is in accord with the vast literature in economics of education, which finds small but significant effects of public investments on educational outcomes in developed and developing countries today.

Conclusion

Historical land settlements had an importance influence on the development of public schooling under the Raj. Areas under the Temporary Settlement where tax assessments were periodically revised received higher land revenues and were able to develop a larger number of public schools charging lower fees. The areas of Bengal and Bihar under the Permanent Settlement of 1793 lacked the same level of public resources because of the fixed nature of their revenues. Consequently, private schools charging higher fees came to represent a larger share of total schools in these provinces. But, the striking differences in public investments had small effects on enrolments and literacy in the colonial period.

The coastal provinces of Bengal, Bombay and Madras had comparable literacy rates averaging 8 to 9 per cent in the early twentieth century despite large differences in public spending between Bengal versus Bombay and Madras. The type of funding, however, may have had second order effects on the distribution of human capital and the subsequent progress of rural literacy. Bengal enjoyed above average English literacy rates in the colonial period. Moreover, differences in literacy between Bengal and Bombay increased over time even though they were similar in the first half of the twentieth century. By 1961 rural literacy in Bombay was twice as high as Bengal. The historical legacy of land revenues may be one factor contributing to this divergence.

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